FIITJEE

ALL INDIA TEST SERIES

PART TEST – I

JEE (Main)-2021

TEST DATE: 05-12-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

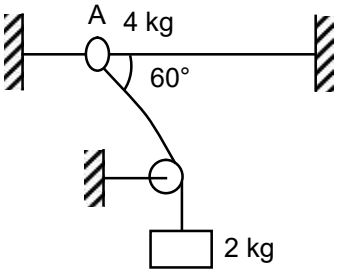
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A smooth ring A of mass 4 kg can slide on a fixed smooth horizontal rod. A string tied to the ring passes over a fixed pulley and carries a block B of mass 2 kg as shown in the figure. At the initial instant, the string attached to the ring makes an angle 60° with the horizontal rod. If the system is released from rest, the initial acceleration of the ring is



(A) $\frac{g}{3}$

(B) $\frac{2g}{3}$

(C) $\frac{2g}{6}$

(D) $\frac{2g}{9}$

2. A uniform disc of m and radius R rolls without slipping down an incline plane of length ℓ and inclination θ. Initially the disc was at rest at the top of the incline plane. Its angular momentum about the point of contact with the inclined plane when it reaches the bottom will be

(A) $\sqrt{\frac{m^2R^2g\ell\sin\theta}{3}}$

(B) $\sqrt{3m^2R^2g\ell\sin\theta}$

(C) $\sqrt{\frac{m^2R^2g\ell\sin\theta}{4}}$

(D) Zero

3. Two bodies of same mass are connected with an inelastic string of length L are placed close together on a horizontal surface. One of them is projected vertically upward from the surface with a velocity $u=\sqrt{6gL}$. The maximum height upto which the centre of mass of the system of two masses will rise is

(A) $\frac{L}{2}$

(B) $\frac{3L}{4}$

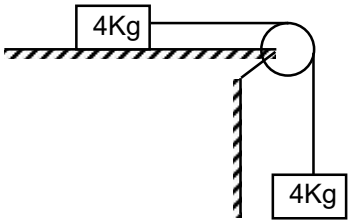
(C) L

(D) $\frac{3L}{2}$

4. A particle is projected along the inner surface of a smooth vertical circle of radius R, with a velocity $\sqrt{\frac{19Rg}{5}}$ at its lowest point. It will leave the circle at an angle from the highest point of the circle is

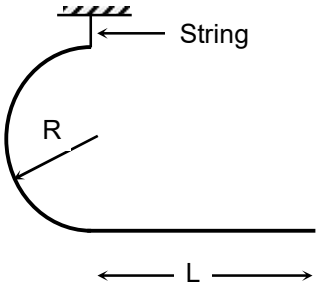
(A) 30°
(B) 37°
(C) 53°
(D) 60°

5. The magnitude of acceleration of centre of mass of the system if coefficient of friction between the block and table is 0.2 and mass of each block is 4 Kg is



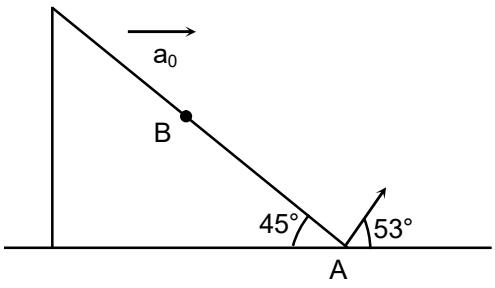
(A) $2\sqrt{2}$ m/s^2
(B) 2 m/s^2
(C) $4\sqrt{2}$ m/s^2
(D) 4 m/s^2

6. An object having uniform linear mass density is suspended in a vertical plane in equilibrium as shown in the diagram. The value of $\frac{R}{L}$ is



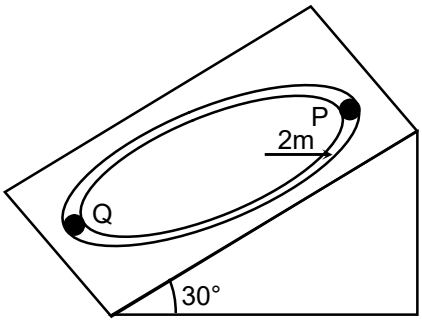
(A) $\frac{1}{2}$
(B) 2
(C) $\frac{1}{2\pi}$
(D) π

7. A particle is projected from point A at an angle of 53° with horizontal. At the same time wedge starts from rest and moves with constant acceleration a_0 as shown in figure. The value of a_0 for which the particle strikes the plane perpendicular to it at B in the frame of wedge is (Take g = 10 m/s^2)



(A) $\frac{130}{15}$ m/s^2
(B) $\frac{140}{15}$ m/s^2
(C) $\frac{150}{15}$ m/s^2
(D) $\frac{160}{15}$ m/s^2

8. A small block of mass 'M' is released from rest, from position P, inside the frictionless circular groove of radius 2m on a fixed inclined plane as shown in figure. The contact force between the block and inclined plane at point Q is



(A) Mg

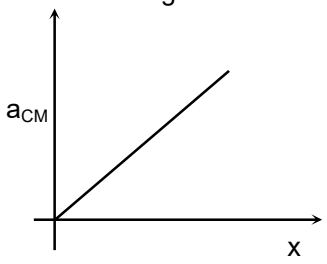
(B) $\sqrt{7}$ Mg

(C) $\frac{\sqrt{7}}{2}$ Mg

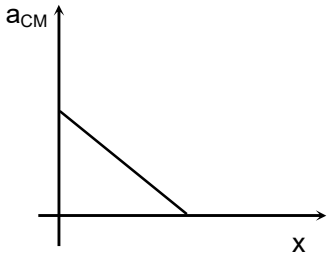
(D) $\frac{\sqrt{3}}{2}$ Mg

9. A rod of mass m and length ℓ is hinged at one of its end P. A Force F is applied at a distance x from P as shown in the figure. The acceleration of centre of mass (a_{CM}) varies with x as;

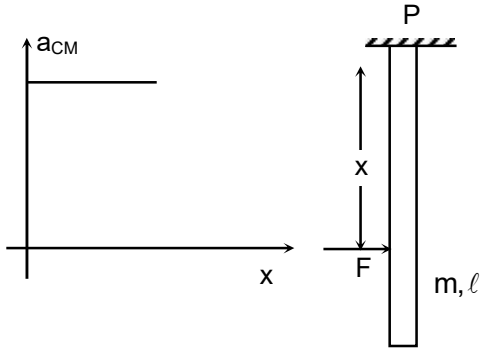
(A)



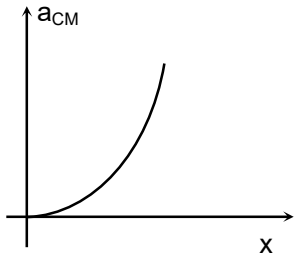
(B)



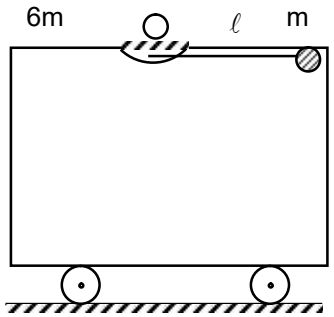
(C)



(D)



10. In the figure shown the cart of mass 6m is initially at rest. A particle of mass m is attached to the end of the light rod which can freely rotate about O. If the rod is released from rest in a horizontal position shown, determine the velocity of the particle with respect to cart when the rod is vertical. Assume friction less surface.



(A) $\sqrt{\frac{14}{3}g\ell}$

(B) $\sqrt{\frac{7}{6}g\ell}$

(C) $\sqrt{\frac{17}{3}g\ell}$

(D) $\sqrt{\frac{7}{3}g\ell}$

11. If velocity vector of a particle just before making an inelastic impact with a smooth horizontal x-y plane surface with coefficient of restitution 0.2 is $\left(15\hat{i}-20\hat{j}-50\hat{k}\right)$ m/s with z-axis being vertically upwards. The velocity vector of the particle just after collision (in m/s) will be

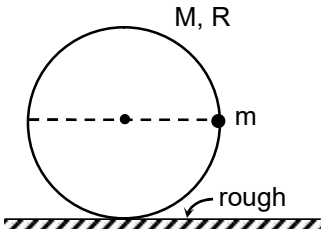
(A) $\left(3\hat{i}-4\hat{j}-10\hat{k}\right)$

(B) $\left(3\hat{i}-4\hat{j}+10\hat{k}\right)$

(C) $\left(15\hat{i}-20\hat{j}+10\hat{k}\right)$

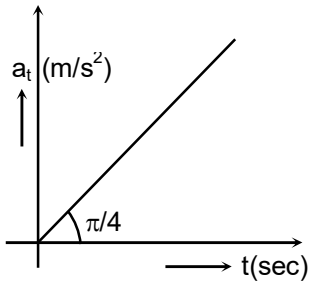
(D) $\left(15\hat{i}-20\hat{j}-10\hat{k}\right)$

12. A particle of mass m = 0.5 kg is rigidly attached to the circumference of a uniform circular disc of mass M = 1 kg and radius R = 20 cm and placed on a rough horizontal surface. The disc is released from rest from the initial position as shown in the figure and it starts rolling without slipping on the rough horizontal surface. The frictional force acting on the disc due to horizontal surface just after release is ($g = 10\ m/s^2$)



(A) 8N
(B) 6N
(C) 5N
(D) 3N

13. Tangential acceleration of a particle starting from rest and moving in a circle of radius 2m varies with time as shown in the graph. Time after which total acceleration of the particle makes an angle of 45° with the radial acceleration is



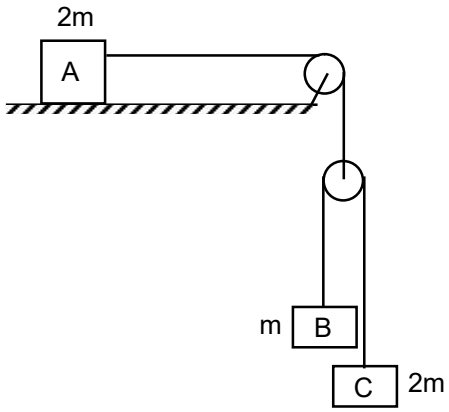
(A) 2 sec

(B) $2\frac{2}{3}\sec$

(C) $2\frac{1}{3}\sec$

(D) $\sqrt{2}\sec$

14. In the given figure if system is released from rest, then find the minimum value of coefficient of friction such that block A of mass 2m does not move



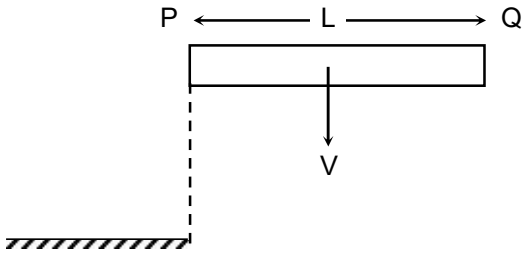
(A) $\frac{1}{3}$

(B) $\frac{2}{3}$

(C) 1

(D) $\frac{4}{3}$

15. A rod PQ of length L and mass M free to move on a frictionless horizontal surface is moving with a velocity v as shown in the figure. End P of the rod PQ strikes the end of wall. Assuming elastic impact, angular velocity of the rod PQ just after impact is



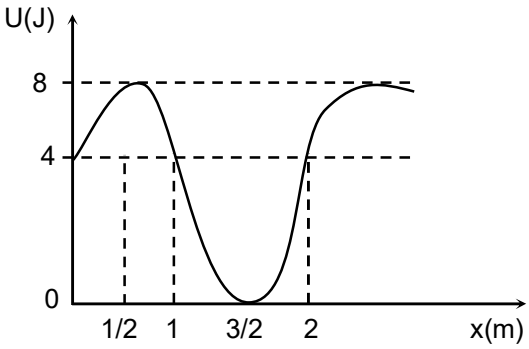
(A) $\frac{v}{2L}$

(B) $\frac{v}{L}$

(C) $\frac{3v}{2L}$

(D) $\frac{3v}{L}$

16. The potential energy curve of a particle varying sinusoidally is shown graphically. Potential energy is varying only in x, it does not depend on y and z coordinates. The maximum value of corresponding conservative force (in magnitude) for range $0 < x < 2m$ (in newton) is



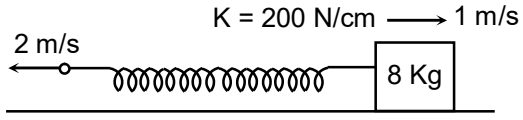
(A) 6.28

(B) 12.56

(C) 9.42

(D) 18.84

17. A spring block system lies on a smooth horizontal surface. The free end of spring is pulled towards left with constant speed 2m/s, the block has a speed 1m/s towards right at t = 0 as shown in the figure. If the spring constant is K = 200 N/cm and mass of the block is 8 kg, the maximum extension of spring is



(A) 2 cm

(B) 4 cm

(C) 6 cm

(D) 8 cm

18. A particle moves along x-axis and its acceleration at any time t is a = 2 sin πt, where t is in second and a in m/s^2. If the particle starts from rest, find the magnitude of displacement (in meter) of the particle from t = 0 to t = t_0 is

(A) $\frac{4}{\pi^2}\sin\pi t_0 - \frac{2t_0}{\pi}$

(B) $\frac{2}{\pi^2}\sin\pi t_0 + \frac{2t_0}{\pi}$

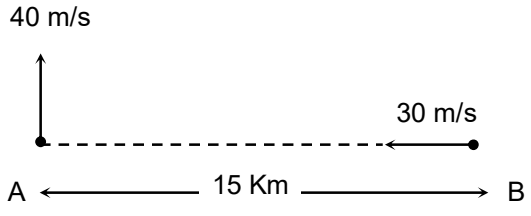
(C) $-\frac{4}{\pi^2}\sin\pi t_0 - \frac{2t_0}{\pi}$

(D) $\frac{-2}{\pi^2}\sin\pi t_0 + \frac{2t_0}{\pi}$

19. Two monkeys A and B are holding on the two sides of a light string passing over a smooth pulley. Mass of A and B are 6 Kg and 10 Kg respectively. 'A' holds the string tightly and 'B' goes down with an acceleration 4 m/s^2 relative to string. Find the weight that A feels of his own body.

(A) 72 N
(B) 60 N
(C) 48 N
(D) Zero

20. Ship 'A' is moving with a velocity 40 m/s due north and ship 'B' is moving with a velocity 30 m/s due west as shown in the figure. After what time the ships will be closest to each other, if the initial separation between them is 15 km?

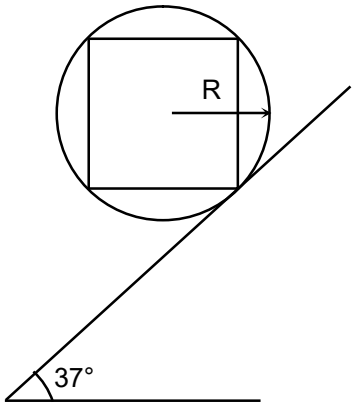


(A) 120 sec
(B) 180 sec
(C) 240 sec
(D) 360 sec

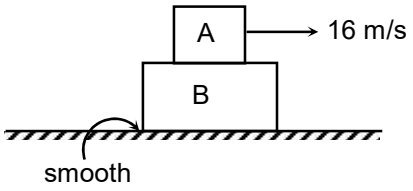
SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. Four identical rods, each of mass m are welded at their ends to form a square, and the corners are then welded to a light metal hoop of radius R. Now the rigid assembly of rods and hoop is allowed to roll down the incline plane as shown in the figure. If the minimum coefficient of static friction which will prevent slipping is $\frac{N}{10}$. Find the value of N.



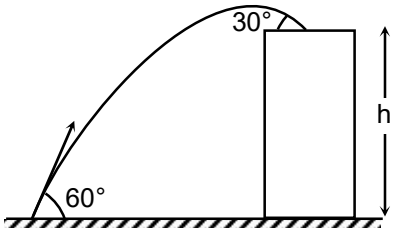
22. A block 'A' of mass 1 Kg is kept on a block 'B' of mass 2 Kg. Block 'A' is imparted a velocity of 16 m/s at t = 0. If the coefficient of friction between A and B is 0.2, find the time (in sec) after which momentum of A and B relative to ground will be equal. (Take g = 10 m/s^2).



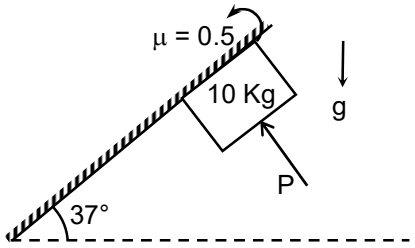
SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

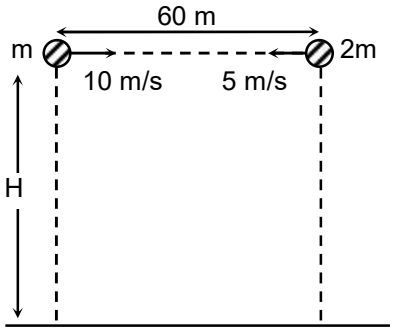
23. A stone is projected at an angle of 60° from the ground level strikes at an angle of 30° on the roof of a building of height 'h'. Find the speed of projection of the stone in m/s. (Take h = 30 m; g = 10 m/s^2)



24. In the figure shown find the minimum value of force P(in N) to be applied perpendicular to the incline so that the block does not slide is (Take g = 10 m/s^2).



25. Two particles one of mass m and the other 2m are projected horizontally towards each other from the same level of height H above the ground with the velocities 10 m/s and 5 m/s respectively at t = 0. They collide in air and stick to each other. If the total time of motion before it strikes the ground is 8 sec. Find the value of H in meter.



Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. Which of the following statements is **not** true about BeH_2?
    (A) It can be obtained by reduction of $BeCl_2$ with $LiAlH_4$
    (B) It is polymeric in nature
    (C) It is a covalent hydride
    (D) It is ionic in nature

27. Choose the correct statement from following:
    (A) H_2 is more rapidly adsorbed on surface for physisorption than D_2
    (B) H_2 is more rapidly adsorbed on surface for chemisorption than D_2
    (C) H_2 reacts slower with Cl_2 than D_2
    (D) H_2 has higher bond dissociation energy than D_2

28. Choose the **incorrect** statement.
    (A) Boron sesquioxide is the anhydride of ortho-boric acid
    (B) H_3BO_3 is a Lewis acid, not a Bronsted acid
    (C) H_3BO_3 no longer used as an antiseptic powder for babies because it sometimes caused rash
    (D) Orthoboric acid is insoluble in H_2O

29. The most basic oxide among the following is
    (A) SiO_2
    (B) MgO
    (C) Na_2O
    (D) Al_2O_3

30. In which of the following process, the magnetic moment value does not change
    (A) $O_2 \rightarrow O_2^-$
    (B) $O_2 \rightarrow O_2^+$
    (C) $N_2 \rightarrow N_2^+$
    (D) $N_2^+ \rightarrow N_2^-$

31. Choose the correct statement:
    (A) If AB_3 molecule is polar then all its bond angles must not be same
    (B) If all the bond angle of AB_3 is same then it must be a planar molecule
    (C) If all bond angle of AB_3 is same then it may or may not be planar molecule
    (D) If only sigma bonds are present then only AB_3 can be a planar molecule

32. Number of distinct spectral line possible during de-excitation, when hydrogen atom in ground state is excited by providing 3.4 eV of energy
    (A) 3
    (B) 0
    (C) 6
    (D) 10

33. For a reaction:

$2A(g) + B(g) \rightleftharpoons C(g) + 3D(g)$

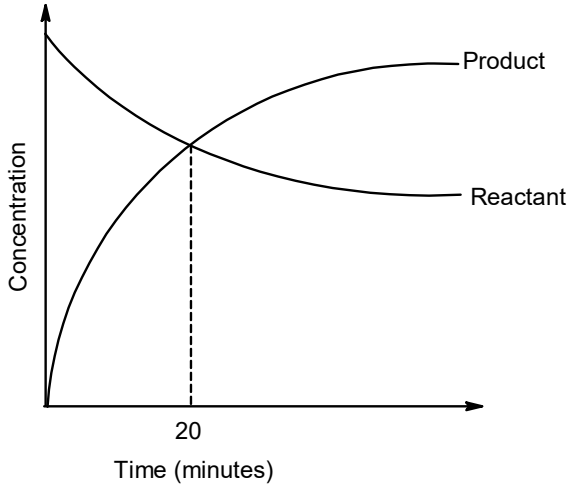
at 300 K, K_c = 2 mol L^{-1}, what must be the volume of the container, if 2 mole each of A, B, C and D are present at equilibrium:

(A) 4 litre
(B) 1 litre
(C) 0.5 litre
(D) 8 litre

34. What will be the concentration of F^- ion in saturated solution of CaF_2? If K_{sp} of CaF_2 is 32×10^{-12} $mol^3\ L^{-3}$

(A) 2×10^{-4} mol/lit
(B) 4×10^{-4} mol/lit
(C) 8×10^{-4} mol/lit
(D) 4×10^{-2} mol/lit

35. For the first order reaction: A $\rightarrow$ 2B, following graph is observed, time required for 75% completion of reaction would be



(A) 40 min
(B) 34.19 min
(C) 68.38 min
(D) 80 min

(Given log 2 = 0.301, log 3 = 0.477)

36. 0.1 mole of $_{92}U^{238}$ undergo radioactive decay to form $_{82}Pb^{206}$ by emitting α and β particles. The number of moles of β - particles emitted in two half life would be

(A) 0.9
(B) 0.45
(C) 1.2
(D) 0.6

37. An element 'X' has IE_1, IE_2 and IE_3 are 738 kJ/mol, 1451 kJ/mol and 7733 kJ/mol respectively. 'X' would be (I.E. = Ionization energy)

(A) Na
(B) Mg
(C) Ne
(D) F

38. The following equation are given:

$\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightleftharpoons NH_3;$ K_1 ....(i)

$2NO \rightleftharpoons N_2 + O_2;$ K_2 ....(ii)

$2H_2 + O_2 \rightleftharpoons 2H_2O;$ K_3 ....(iii)

What will be equilibrium constant (K) for following reaction:

$2NO + 3H_2O \rightleftharpoons 2NH_3 + \frac{5}{2}O_2;$ K ...(iv)

(A) $K_1^2.K_2.K_3^{3/2}$

(B) $K_2.K_1^{1/2}.K_3^{3/2}$

(C) $\frac{K_1^2.K_2}{K_3^{3/2}}$

(D) $\frac{K_3^{3/2}.K_2}{K_1^2}$

39. Which of the following solution cannot act as buffer solution when equal volume of each is mixed?

(A) 0.1 M CH_3COONa + 0.12 M CH_3COOH

(B) 0.2 M CH_3COONa + 0.1 M HCl

(C) 0.1 M CH_3COONa + 0.2 M HCl

(D) 0.1 M NH_3 + 0.1 M CH_3COOH

40. For the following reaction:

$BaCl_2 + Na_3PO_4 \longrightarrow Ba_3(PO_4)_2 + NaCl$

Minimum number of moles of $BaCl_2$ required for complete consumption of 1 mole of Na_3PO_4

(A) $\frac{2}{3}$

(B) $\frac{3}{2}$

(C) 2

(D) 1

41. Choose the correct option for following statements:

(I) CaC_2 on hydrolysis produce acetylene

(II) Al_4C_3 on hydrolysis produces propyne

(III) $Al_2(CH_3)_6$ has 3 – centre, 2 – electron bond

(IV) Al_4C_3 on hydrolysis produces methane

(A) I, II and III are correct statements

(B) only I and IV statements are correct

(C) I, III and IV statements are correct

(D) II and III statements are incorrect

42. Choose **incorrect** statement:

(A) S has a lower electron gain enthalpy than Cl

(B) B has lower ionization energy than Be

(C) Cl has lower ionization energy than F

(D) P has lower ionization energy than S

43. Which of the following pair are isotones?

(A) ${}_6C^{14}$ and ${}_7N^{14}$

(B) ${}_{32}Ge^{76}$ and ${}_{33}As^{77}$

(C) $_{26}Fe^{58}$ and $_{27}Ni^{58}$

(D) $_{16}S^{32}$ and $_{16}S^{34}$

44. Temperature coefficient for a first order reaction is 2. If the half life for the reaction at 25°C is 10 min, what would be half life of same reaction at 45°C?
(A) 20 min
(B) 5 min
(C) 150 seconds
(D) 120 seconds

45. For the following equilibrium reaction:

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

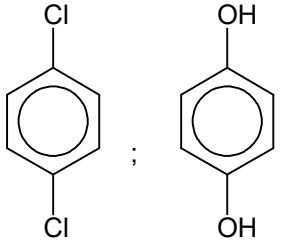
$CH_3OH(g)$ will be consumed when
(A) pressure of the system is increase
(B) volume of the system decreased
(C) Ar gas is added of constant volume
(D) Ar gas is added at constant pressure

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. The ratio of wave numbers associated with longest wave length of Paschen series for Li^{2+} ion to the shortest wavelength of Brackett series of H – atom would be.

47. How many of the following molecules have zero dipole moment:
$CHCl_3$, CH_2Cl_2, SF_4, SF_6, BF_3, NH_3, PCl_3F_2, PCl_2F_3



SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. For the following reaction:

$$Cu + HNO_3 \longrightarrow Cu(NO_3)_2 + NO_2 + H_2O$$

What will be the ratio of stoichiometric coefficient of HNO_3 to NO_2 in balanced reaction?

49. Acetic acid (100 ml, 0.1 M) is titrated with 0.1 M NaOH. What will be the difference in pOH reading, when 80 ml NaOH is added and when 100 ml NaOH is added?
(Given pK_a (CH_3COOH) = 5, log 2 = 0.3)

50. Equal volume of 0.1 M weak acid HA (K_b of A^- = 5×10^{-10}) and NaOH (0.1 M) is mixed, what will be the percentage of hydrolysis of NaA formed at 298 K?

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. There are three events A, B, C such that n(B) = 230, $n(\bar{A} \cap B \cap \bar{C}) = 100$, $n(A \cap B \cap \bar{C}) = 50$ and $n(A \cap B \cap C) = 10$, then value of $n(\bar{A} \cap B \cap C)$ is
(A) 30
(B) 40
(C) 50
(D) 70

52. The value of $\lim\limits_{t\to\infty} \dfrac{\int_{1/3}^{t} [\cot^{-1} x]\,dx}{\int_{1/2}^{t} \left[\dfrac{x+1}{x}\right] dx}$ where [.] denotes the greatest integer function, is equal to
(A) $\cot 1 - \dfrac{1}{3}$
(B) cot 1
(C) 0
(D) $\cot 1 + \dfrac{1}{2}$

53. If f(x) = |3 – 2x|, then domain of the function $\sin^{-1}(f(|x|))$ is
(A) $[-3, -1] \cup [1, 3]$
(B) $[-2, 2]$
(C) $[-3, -2] \cup [2, 3]$
(D) $[-2, -1] \cup [1, 2]$

54. Let f(x) = max{$|x^2 - 3|x||$, |x|} and g(x) = min{$|x^2 - 3|x||$, |x|}, then
(A) f(x) and g(x) not differentiable at 5 and 7 points respectively
(B) f(x) and g(x) not differentiable at 7 and 5 points respectively
(C) f(x) and g(x) not differentiable at 5 and 6 points respectively
(D) f(x) and g(x) not differentiable at 6 and 5 points respectively

55. Suppose $f(x) = (3)^{px} + (3)^{qx}$, where $p \neq q$ and $\dfrac{f''(x)}{(\log_e 3)^2} - \dfrac{2f'(x)}{(\log_e 3)} - 15f(x) = 0$ for all x, then the value of |p + q| is
(A) 0
(B) 1
(C) 2
(D) 3

56. If $\frac{12a+5b}{15} = -\frac{(c+d)}{4}$, then the equation $ax^4 + bx^2 + cx + d = 0$ has
(A) no real solution
(B) at least one real root in (–2, 0)
(C) at least one real root in (0, 2)
(D) at least one real root in (–2, –1)

57. The graphs $y - 2 = 2x(x^2 - 2)$ and $y + 1 = x(x^2 + 2)$ intersect at exactly 3 distinct points. If all the three points are collinear. Then slope of the line is
(A) 2
(B) 4
(C) 6
(D) 8

58. If p(x) is polynomial satisfying $p(x^3) = x^2\, p(x^2)$ and $p(0) = -3$, $p'\left(\frac{1}{3}\right) = 0$, $p(1) = 2$. Then p(3) is equal to
(A) 101
(B) 102
(C) 103
(D) 104

59. If $\lambda = \sum_{n=1}^{\infty} \frac{(12)^n}{(4^n - 3^n)(4^{n+1} - 3^{n+1})}$, then value of λ is equal to
(A) 1
(B) 2
(C) 3
(D) none of these

60. Let f(x) be continuous and differentiable function for all real x and $f(x + y) = f(x) - xy + f(y)$ if $\lim_{h\to 0} \frac{f(h)}{h} = 2$, then f(5) – f(3) is
(A) –2
(B) 2
(C) 4
(D) –4

61. If the equation $ax^n + 3x^5 + bx^2 + c = 0$, $n \in I^+$ has infinite number of real solutions. Then a + b + c + n is equal to
(A) 2
(B) 3
(C) 4
(D) 6

62. Solution of $\int \frac{9\cos 2x + 12\cos x - 6\sin x}{(2+3\sin x)(4+3\cos x)} dx$
(A) $\log_e\left(\frac{2+3\sin x}{4+3\cos x}\right) + c$
(B) $\log_e\left(\frac{4+3\cos x}{2+3\sin x}\right) + c$
(C) $\log_e((2+3\sin x)(4+3\cos x)) + c$
(D) $-\log_e(2+3\sin x)(4+3\cos x) + c$

63. Solution of $\int \frac{x^4+3}{\sqrt{x^8+x^6-3x^4}}dx$ is

(A) $\frac{\sqrt{x^4+x^2-3}}{x}+c$

(B) $-\frac{\sqrt{x^4+x^2-3}}{x}+c$

(C) $\frac{x}{\sqrt{x^4+x^2-3}}+c$

(D) $-\frac{x}{\sqrt{x^4+x^2-3}}+c$

64. The solution of $\int x^x \cdot \log_e x \cdot \left(\log_e (ex)^x\right)dx$ is

(A) $ex^x \cdot \log_e\left(\frac{e}{x^x}\right)+c$

(B) $x^x \log_e\left(\frac{e}{x^x}\right)+c$

(C) $x^x \log_e\left(\frac{x^x}{e}\right)+c$

(D) $ex^x \log_e\left(\frac{x^x}{e}\right)+c$

65. Let the function f satisfies $f(2x)\cdot f'(-2x) = f(-2x)f'(2x)$ and $f(0) = 4$, then $\int_{-24}^{24} \frac{dx}{4+f(2x)}dx$ is

(A) 4
(B) 6
(C) 8
(D) 12

66. The value of $\int_0^{\pi/8} \left[\sec x + 2\left[\tan x + 3\left[\cos x + 4[\sin x]\right]\right]\right]dx$ (where [.] denote greatest integer function), is

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{6}$

(C) $\frac{\pi}{8}$

(D) $\frac{\pi}{10}$

67. Let 'b' be a positive constant. Let two curves $S_1 : e^{-x}y = 1$ and $S_2 : e^{x}y = e^{b}$. Let A be the area of the part surrounding by S_1, S_2 and x = 0, then $\lim\limits_{b\to 0} \frac{3A}{4b^2}$ is equal to

(A) $\frac{1}{4}$

(B) $\frac{3}{16}$

(C) $\frac{3}{17}$

(D) $\frac{2}{19}$

68. Let set of curves $P_i : (y-i)^2 = (x-i)$ and $Q_i : (x-i)^2 = (y-i)$ where i = 0, 1, 2, ….. n and A_i is the area enclosed by curve P_i and Q_i, then the value of $\sum\limits_{i=0}^{20} A_i$

(A) 7 sq. units
(B) 14 sq. units
(C) 21 sq. units
(D) 11 sq. units

69. Let R be the relation on the set R of all real numbers defined by a R b iff $|a - b| \le 1$, then R is
(A) reflexive and symmetric
(B) equivalence
(C) transitive and reflexive
(D) anti-symmetric, reflexive and symmetric

70. Consider the following two statements
$S_1 : (p \vee q) \wedge (\sim p \wedge \sim q)$ is contradiction
$S_2 : (p \wedge q) \vee \sim p = \sim p \vee q$, then S_1 and S_2 are respectively
(A) T, T
(B) T, F
(C) F, T
(D) F, F

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. A real valued function f(x), $f : \left(0, \frac{\pi}{2}\right) \to R^+$ satisfies the differential equation $f^2(x) = \frac{f(x) + x f'(x) - 1}{x^2}$ and $f\left(\frac{\pi}{4}\right) = \frac{4}{\pi}$, then the value of $\lim\limits_{x\to 0} \left(1 + f(x) + f^2(x) + f^3(x)\right)$ is

72. If g(x) is the inverse of f(x) and f(x) has domain $x \in [2, 7]$ where f(2) = 3 and f(7) = 13, then the value of $\frac{1}{17}\left(\int_2^7 f(x)\,dx + \int_3^{13} g(y)\,dy\right)$ is equal to

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. A curve passing through (2, 3) and satisfying $\int_0^x tf(t)dt = x^2 f(x)$, (x > 0). Then the value of f(6) + f(8) + f(10) is

74. Let $a = (7)^{\frac{1}{25}} + 1$ and for all $n \geq 3$.
Let $f(n) = {}^nC_0(a)^{n-1} - {}^nC_1(a)^{n-2} + {}^nC_2(a)^{n-3} + ... + (-1)^{n-1}\ {}^nC_{n-1}(a)^0$ if the value of f(2020) + f(2021) = $(7)^{\lambda}$, then the value of λ is

75. The area enclosed by the figure described by the equation $x^4 + 1 = 2x^2 + y^2$ is